

Omar Elmasaoudi

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EDUCATION

Concordia University

Montreal, QC

Bachelor of Engineering in Software Engineering

2022-2026

- Relevant course-work: Databases, Data structures and Algorithms, OOP (Java), Web programming (php, js), Functional Programming (Erlang, Clojure), Software Processes and lifecycle, & Software Architecture

WORK EXPERIENCE

Air Ocean Maroc

Rabat, Morocco

Software Developer Intern

April 2024 – August 2024

- Collaborated with DevOps engineers to deploy a Medivac software system to the company's network, ensuring seamless integration and scalability.
- Worked closely with medics and operations personnel to verify that all requirements were met, addressing real-time scheduling, supply tracking, and secure access to patient and personnel data.
- Enhanced flight and ground Medivac operations efficiency by 40% through the implementation of key features such as real-time scheduling and supply tracking.

PROJECTS

Stock Performance and Analysis Tool | *Python, NextJs, Typescript, MongoDB*

November 2024 – Present

- Developing a portfolio management tool that uses **NLP** to summarize stock-related articles and provide actionable insights for users, such as forecasting stock prices and guiding decisions on keeping, selling, or shorting stocks.
- Leveraging **pre-trained language models (LMs)** and applying **transfer learning** to fine-tune them for analyzing sentiment in financial articles from sources like Google Finance, Bloomberg, and Yahoo Finance. Building a **Neo4j graph database** with scalar relationship values, dynamically calculated through a deep neural network, to visualize company connections and their impact on stock performance.
- Implementing the frontend using **Next.js**, a **TypeScript** backend for handling user data, and a Python-based **web scraper** to label large datasets for training the models and improving data representation.

Managed Medivac Application | *React, Typescript, PostgreSQL, Docker*

February 2024 – April 2024

- Enhanced medivac operations with a centralized platform for stock, personnel, and patient data. Features include supply tracking, scheduling, real-time flight updates, and secure access – boosting efficiency and response times.
- Designed a responsive UI with **React**, **Vite**, and Chakra UI for optimal user experience.
- Implemented a scalable and fault-tolerant **microservices** architecture for flexibility.
- Secured the platform with a custom **TypeScript** ID provider, NGINX reverse proxy, and **JWT**-based authentication.
- Automated AWS deployment with a custom CI/CD pipeline (Docker) for rapid updates.

Operations Manager Application | *JavaScript, express-JS, React, MySQL*

May 2023 – August 2023

- Developed a comprehensive management program for orders, medical/maintenance stock, and drug inventory – streamlining operations and optimizing decision-making.
- Built a full-stack web application (**JavaScript, Express.js, React**) with a REST API for centralized operations data.
- Integrated inventory monitoring to ensure efficient resource allocation, preventing shortages and overstocking.
- Secured sensitive data with **JWT**-based user authentication and authorization.
- Deployed on **AWS** for scalability and rapid development, then transitioned seamlessly to company servers for long-term use.
- Designed a relational database schema using **MySQL** to optimize relationships between orders, inventory, history, and missions, enhancing operational efficiency.

Java Snakes and Ladders Game | *Java, Java Swing*

September 2023 – September 2023

- Designed and implemented a **GUI-based Snakes and Ladders game** using **Java Swing**, featuring a multi-class architecture to manage gameplay, player interactions, and GUI components.

CERTIFICATIONS

DeepLearning.AI Machine Learning Specialization

May 2024 – July 2024

- Learned foundational concepts of supervised learning, including **linear regression**, **logistic regression**, **neural networks**, and **decision trees**. Gained expertise in unsupervised learning techniques like **clustering** and **anomaly detection** to analyze and model unlabeled data.
- Applied advanced techniques to build **recommender systems**, leveraging **collaborative filtering** and **content-based methods**. Explored practical applications of deep learning with projects on **CNNs**, **optimization algorithms**, and **reinforcement learning**.

DeepLearning.AI Natural Language Processing (NLP) Specialization

August 2024 – November 2024

- Mastered techniques including **logistic regression**, **naïve Bayes**, and **word vectors** to implement sentiment analysis, complete analogies, and translate words. Used **dynamic programming**, **hidden Markov models**, and **word embeddings** to build autocorrect, autocomplete, and identify part-of-speech tags for words.
- Developed advanced models like **recurrent neural networks (RNNs)**, **LSTMs**, **GRUs**, and **Siamese networks** for sentiment analysis, text generation, and named entity recognition. Applied **encoder-decoder models**, **causal**, and **self-attention mechanisms** for tasks like machine translation, text summarization, and question answering.

DeepLearning.AI Convolutional Neural Networks (CNN) Course

July 2024 – August 2024

- Built **convolutional neural networks (CNNs)**, including advanced models like **Residual Networks (ResNets)**, **MobileNet**, and **EfficientNet**. Applied CNNs to tasks such as **object detection**, **recognition**, and **landmark detection**, leveraging techniques like **YOLO** and **U-Net** for **semantic segmentation** and **bounding box predictions**.
- Explored advanced applications of CNNs, including **face recognition** using **one-shot learning**, **Siamese networks**, and **triplet loss**. Developed **neural style transfer** models to generate art and applied convolutional techniques to **image**, **video**, and **3D data** through **transfer learning** and **data augmentation**.

DeepLearning.AI TensorFlow Developer Professional Certificate

August 2024 – Present

- Learned best practices for using **TensorFlow** to train neural networks for computer vision applications. Handled real-world image data and implemented strategies to prevent overfitting, including **data augmentation** and **dropout**.
- Built natural language processing systems using **TensorFlow**, applying **RNNs**, **GRUs**, and **LSTMs** to train models on text repositories for tasks such as text generation and sentiment analysis.

Docker and Kubernetes Complete Guide | Udemy

February 2024 - March 2024

- Mastered containerization concepts with **Docker**, including building, deploying, and managing containerized applications, and leveraging Docker Compose for multi-container environments.
- Learned orchestration with **Kubernetes**, including setting up clusters, managing deployments, scaling applications, and configuring services using best practices for high availability and fault tolerance.

HACKATHONS

@Hack CTF

March 3, 2024

- Participated in my first cybersecurity-focused CTF competition, collaborating with teammates to tackle various challenges in a 24-hour setting.

SKILLS

Programming Languages: Java, Python, Typescript

Frameworks: TensorFlow, Numpy, Pandas, NodeJs, Nextjs, Tailwind CSS

Databases: Postgres, Mysql, Neo4j, MongoDB, Redis

Developer Tools: Git, AWS, Docker, Vercel

Languages: English (Native), French (Fluent), Arabic (intermediate)